

PYTHON PROJECT · CM25013

Address Book in Python

A menu-driven contact management application built with core Python

PRESENTED BY

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Course Code: CM25013

Introduction



What it is

A Python application for managing personal and professional contacts.



Core actions

Users can add, view, search, update, and delete contact records.



Interface

A simple, menu-driven interface keeps interaction easy for any user.



Contact fields

Each entry can include name, phone number, email, postal address, and more.



Persistence

Records are stored permanently using file handling or a database.

Objectives

1

Create a simple and user-friendly contact management system

2

Store contact details in an organized manner

3

Provide fast searching and viewing of contacts

4

Allow users to modify existing contact information

5

Provide a delete option for outdated contacts

6

Maintain data permanently using file handling or a database

Main Features



Add Contact

Save a new contact with required details.



View Contacts

Display all saved contacts at a glance.



Search Contact

Find a contact using name or phone number.



Update Contact

Edit existing contact information.



Delete Contact

Remove a selected contact from records.



Permanent Storage

Keep records safe between program runs.

Technologies Used

1

Python

Core programming language

2

Functions

Separate operations into reusable blocks

3

Lists / Dictionaries

Represent and organize contact data

4

File Handling

Store records permanently

5

Conditional Statements

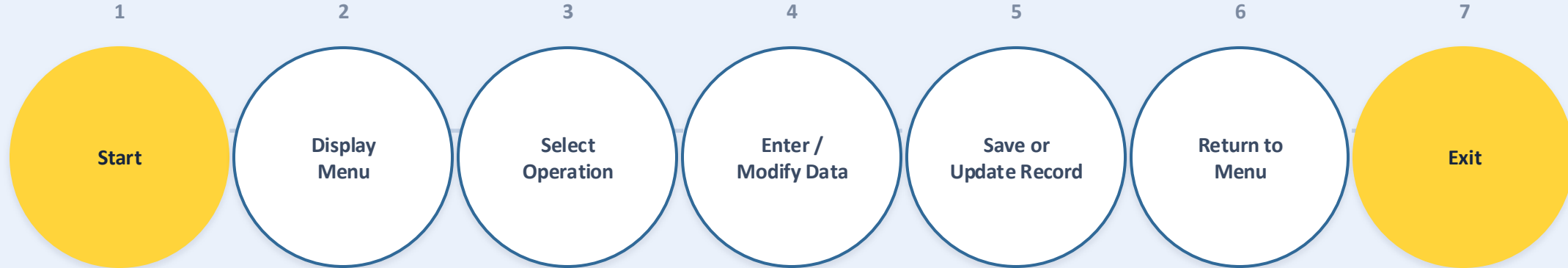
Control menu choices and validation

6

Loops

Repeat the menu until the user exits

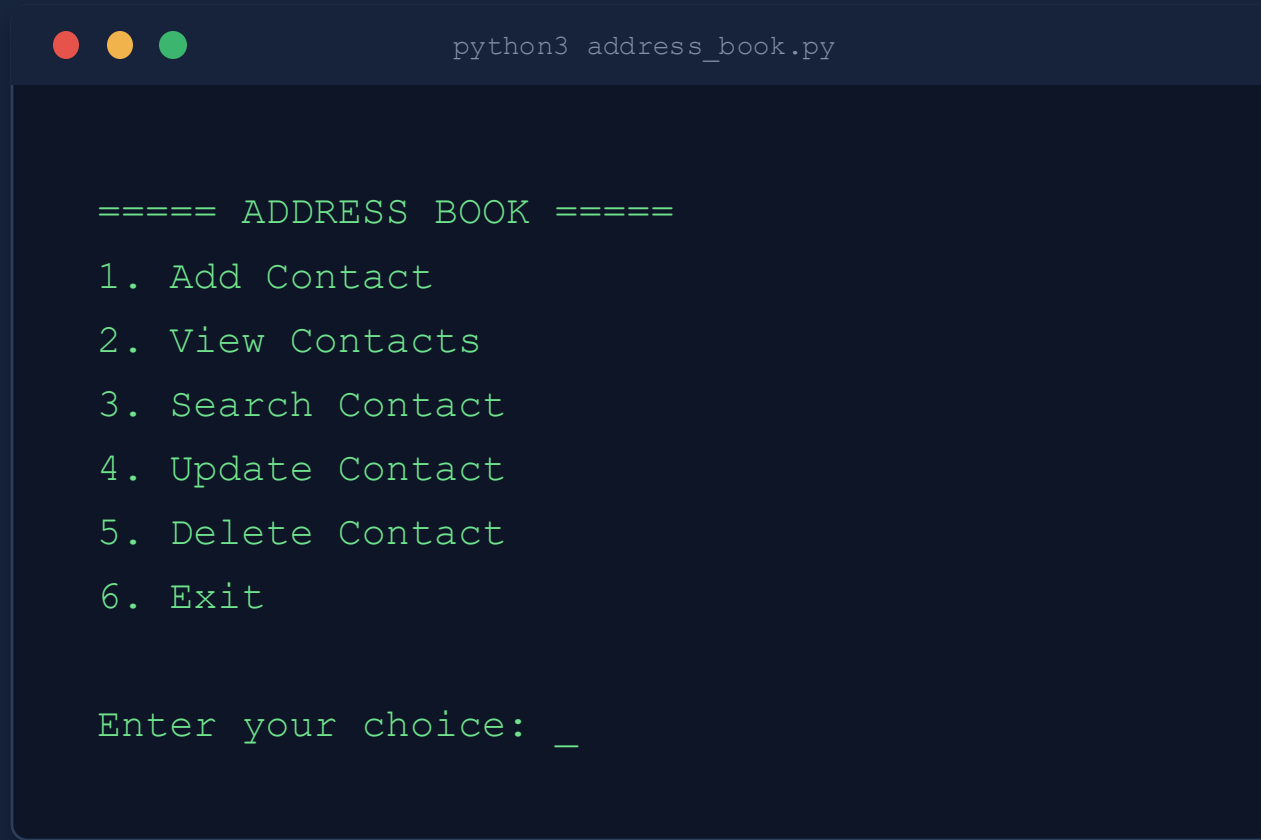
Working Flow



The loop keeps going

After each operation the program returns to the main menu, so the cycle repeats until the user chooses to exit.

Sample Menu



```
python3 address_book.py

===== ADDRESS BOOK =====
1. Add Contact
2. View Contacts
3. Search Contact
4. Update Contact
5. Delete Contact
6. Exit

Enter your choice: _
```

Data Structure

Each contact is represented using a Python dictionary. Multiple contact dictionaries are stored in a list, and persisted in a file or database.

name

→ Contact name

phone

→ Mobile / telephone number

email

→ Email address

address

→ Postal address

```
contact = {  
    "name": "Ravi Sharma",  
    "phone": "9876543210",  
    "email": "ravi@mail.com",  
    "address": "Pune, India"  
}  
  
address_book = []  
address_book.append(contact)
```

Advantages & Future Scope

ADVANTAGES

- Easy to use and suitable for beginners
- Reduces manual effort in maintaining contact records
- Search and update operations make contact management faster
- File/database storage helps preserve records permanently

FUTURE SCOPE

Graphical User Interface

A GUI for a more visual, user-friendly experience

Login System

Secure access with user authentication

Contact Groups

Organize contacts into custom categories

Import / Export

Move contact data in and out easily

Database Integration

Scale storage with a full database backend

CONCLUSION

A simple, efficient way to store, organize, search, update, and delete contact information.

Thank You

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